5.2 Absolute maximum ratings

Stresses above the absolute maximum ratings listed in *Table 21*, *Table 22* and *Table 23* may cause permanent damage to the device. These are stress ratings only and functional operation of the device at these conditions is not implied. Exposure to maximum rating conditions for extended periods may affect device reliability. The device mission profile (application conditions) is compliant with the JEDEC JESD47 qualification standard.

All voltages are defined with respect to V_{SS}.

Table 21. Voltage characteristics

Symbol	Ratings	Min	Max	Unit
V_{DD}	External supply voltage	- 0.3	4.0	V
V_{REF+}	External voltage on VREF+ pin	- 0.3	Min(V_{DD} + 0.4, 4.0)	V
V_{IN}(1)	Input voltage on pin	- 0.3	V_{DDIO1} + 4.0(2)(3)	V

1. V_{IN} maximum must always be respected. Refer to *Table 22* for the maximum allowed injected current values.
2. To sustain a voltage higher than 4 V the internal pull-up/pull-down resistors must be disabled.
3. When an FT_a pin is used by an analog peripheral such as ADC, the maximum V_{IN} is 4 V.

Table 22. Current characteristics

Symbol	Ratings	Max	Unit
$I_{VDD/VDDA}$	Current into VDD/VDDA power pin (source)	100	mA
$I_{VSS/VSSA}$	Current out of VSS/VSSA ground pin (sink)	100	mA
$I_{IO(PIN)}$	Output current sunk by any I/O and control pin	20	mA
	Output current sourced by any I/O and control pin	20	
$\sum I_{(PIN)}$	Total output current sunk by sum of all I/Os and control pins(1)	80	mA
	Total output current sourced by sum of all I/Os and control pins(1)	80	
$I_{INJ(PIN)}$(1)(2)	Injected current on a FT_xx pin	-5 / NA	mA
$\sum I_{INJ(PIN)}$	Total injected current (sum of all I/Os and control pins)(3)	-25	mA

1. Positive injection is not possible on these I/Os and does not occur for input voltages lower than the specified maximum value.
2. A positive injection is induced by $V_{IN} > V_{DDIO1}$ while a negative injection is induced by $V_{IN} < V_{SS}$. $I_{INJ(PIN)}$ must never be exceeded. Refer also to *Table 21: Voltage characteristics* for the maximum allowed input voltage values.
3. When several inputs are submitted to a current injection, the maximum $\sum|I_{INJ(PIN)}|$ is the absolute sum of the negative injected currents (instantaneous values).

Table 23. Thermal characteristics

Symbol	Ratings	Value	Unit
T_{STG}	Storage temperature range	–65 to +150	°C
T_J	Maximum junction temperature	130	°C